

International Conference on Machine Learning and Data Engineering

QAHNet: A Cross-Domain Framework for Low-Quality Image Enhancement and Adaptive Recognition in Multi-Task Disease Diagnosis

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Abstract

Disease diagnosis from mobile-acquired images in real-world settings remains a critical challenge, as uncontrolled capture conditions introduce noise, blur, and resolution loss that cause standard deep learning models to degrade sharply in classification accuracy across human and plant disease domains. Existing approaches address this problem either through standalone image enhancement pipelines or single-domain classification models that are trained on clean, controlled imagery and fail to generalize across domains or jointly predict disease severity alongside category. We propose a cross-domain multi-task framework that performs feature-space quality adaptation via Feature-wise Linear Modulation (FiLM), enabling a shared hybrid CNN-Transformer architecture to dynamically adjust its internal representations to each input's degradation level while jointly training disease classification and severity grading heads across human and plant disease imagery. The framework is trained and evaluated on the PAD-UFES-20 dermatological dataset and the PlantVillage plant disease dataset, with a stochastic degradation simulator applying Gaussian blur, additive noise, and JPEG compression on-the-fly during training to emulate realistic mobile imaging conditions. The proposed framework achieves a validation AUC of 0.876 on disease classification and a weighted severity F1-score of 0.705, outperforming CNN-only and plain CNN-Transformer baselines across all four degradation conditions tested, demonstrating robust and generalizable diagnostic performance suitable for real-world deployment at scale.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering.

Keywords: cross-domain disease diagnosis; low-quality image classification; multi-task learning; severity assessment; feature-wise linear modulation; mobile imaging; image degradation robustness; transfer learning; explainability; deep learning

1. Introduction

The use of deep learning for disease diagnosis has grown rapidly in recent years, with models now capable of identifying diseases from images across domains such as dermatology, radiology, and agriculture [1]. However, most

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of these models are developed and tested on high-quality, controlled images captured under ideal conditions. In practice, images are often taken using mobile phones in outdoor or clinical settings where lighting is poor, the camera may move during capture, and the resolution is low [3]. These real-world conditions introduce noise, blur, and other distortions that cause even well-trained models to perform poorly [4]. This gap between how models are trained and how they are actually used is one of the biggest barriers to deploying diagnostic systems in the real world.

Several methods have been proposed to deal with image quality issues, including image enhancement techniques such as super-resolution and denoising that are applied before the image is passed to the classifier [16]. Others have explored building more robust classifiers using data augmentation or domain adaptation [13]. However, these solutions are typically designed for a single disease domain, such as only skin diseases or only plant diseases, and they rarely combine image quality handling with multi-task learning [10, 9]. Most existing models also only predict whether a disease is present or what type it is, without estimating how severe it is, even though severity information is often more useful for treatment decisions [11, 12]. There is therefore a clear need for a unified framework that can handle degraded images, work across different disease domains, and jointly predict both disease category and severity.

This paper proposes a cross-domain multi-task framework that addresses all three of these limitations together. The framework uses a hybrid CNN-Transformer architecture where a pretrained EfficientNet-B0 backbone extracts image features, which are then passed to a transformer encoder that is conditioned on an estimated image quality score using Feature-wise Linear Modulation (FiLM) [28, 18]. This quality conditioning allows the model to adapt its internal processing depending on how degraded the input image is, rather than treating all images the same way. During training, synthetic degradations including Gaussian blur, additive noise, and JPEG compression are applied on-the-fly to simulate mobile imaging conditions [3, 6]. The model is trained jointly on the PAD-UFES-20 dermatological dataset and the PlantVillage plant disease dataset, with separate prediction heads for disease classification and severity grading [27, 8]. Attention-based explainability is also incorporated so that the regions driving each prediction can be visualized, making the model more transparent and trustworthy for clinical and agricultural use [17]. Experiments show that the proposed framework achieves a validation AUC of 0.876 for disease classification and a weighted severity F1-score of 0.705, outperforming baseline models under all degradation conditions tested.

2. Related Work

2.1. Deep Learning for Disease Classification

Deep learning has been widely adopted across diagnostic domains because of its ability to learn discriminative features directly from raw images. A systematic review comparing deep learning against traditional machine learning for lung disease diagnosis found that automatic hierarchical feature extraction consistently outperforms handcrafted pipelines [1]. A comprehensive review on tuberculosis detection from chest radiographs benchmarked state-of-the-art models for low-resource clinical settings [2]. CNNs applied to Alzheimer's disease classification eliminated handcrafted features while improving diagnostic efficiency, though performance remained tied to dataset quality [27]. A broader systematic review covering MRI, ultrasound, and mammography found that deep learning pipelines handle breast cancer classification comprehensively but that heterogeneity across studies limits direct comparison [14]. An attention-based hybrid framework integrating brain connectivity and activity from resting-state fMRI demonstrated that multimodal fusion improves psychiatric disorder diagnosis, though at the cost of interpretability and large data requirements [15]. Transfer learning on pre-trained CNNs has also shown effectiveness for skin lesion classification [26] and emerging diseases such as monkeypox, where data scarcity is a major concern [10]. Skin cancer detection using combined deep learner decisions has further demonstrated strong classification performance across multiple lesion types [23]. Collectively, these works confirm the diagnostic value of deep learning but remain confined to single domains and clean imaging conditions.

2.2. Hybrid CNN-Transformer Architectures

To improve representational capacity, recent work has explored hybrid CNN-Transformer designs that combine local feature extraction from CNNs with the global context modeling of transformer architectures [21]. A C-Swin model combining CNN local features with Swin Transformer global attention achieved approximately 96% accu-

racy for lung cancer classification but risks overfitting on small datasets and carries high computational cost [28]. A survey of over 170 Transformer-based models highlights Vision Transformers and self-attention for multi-modal medical imaging while noting high computational cost and limited interpretability [17]. CTGFSR combined dual-branch CNN-Transformer fusion with gated attention for medical image super-resolution, improving PSNR and SSIM without additional scans [16]. A review of CNN- and Transformer-based segmentation architectures documented the shift toward long-range dependency modeling while flagging deployment scalability as an open problem [29, 25]. TransMed fused CNN and Transformer components for multi-modal MRI classification, improving accuracy by approximately 10 percentage points over single-modality baselines [18]. Recent Vision Transformer architectures have also incorporated patient metadata alongside image features for skin lesion classification, demonstrating that metadata fusion further improves diagnostic performance [22]. While these hybrid designs show clear accuracy gains, they consistently trade off interpretability and computational efficiency, limiting their use in real-world settings.

2.3. Image Quality and Degradation Robustness

Controlled perturbation studies on radiographic data show that resolution and blur are the dominant factors degrading fracture-detection performance [4], and standard classifiers lose substantial accuracy under degradation relative to human perception [3]. Quality-aware enhancement has improved infant retinopathy screening in fundus imaging [5], while deep learning models trained on smartphone-acquired pathology images have demonstrated that reliable disease diagnosis is feasible even under mobile imaging conditions [7]. Ensemble CNNs with Grad-CAM interpretability have further classified tea leaf disease under low-resolution field conditions [6]. Multi-task frameworks using triple-branch Swin Transformers have jointly identified plant disease type and severity [11], and attention-based frameworks have performed joint disease detection and severity grading in veterinary imaging [12]. However, these works address image quality and multi-task learning in isolation and do not combine both within a cross-domain setting.

2.4. Research Gaps

Despite the progress reviewed above, several important gaps remain unaddressed. Most models are trained on clean, high-resolution images and fail to account for degradation from mobile phone capture in real-world settings [3, 4]. Existing frameworks are almost always single-domain, meaning a model built for skin disease cannot be applied to plant disease even though the core image quality challenges are the same [10, 9]. Most systems only predict disease category without estimating severity, limiting their usefulness for clinical or agricultural decision-making [11, 12]. Hybrid CNN-Transformer models tend to be computationally heavy and difficult to interpret, making deployment in resource-constrained settings challenging [28, 18]. To address these gaps, this paper proposes a cross-domain multi-task framework that handles degraded mobile-acquired images across human and plant disease domains, jointly predicts disease category and severity, incorporating lightweight attention-based explainability using FiLM conditioning to dynamically adapt to each input's degradation level.

3. Proposed Methodology

To address the limitations of existing disease diagnosis systems, this paper proposes a cross-domain multi-task framework specifically designed to improve diagnostic performance under real-world conditions where image quality varies significantly. The framework combines CNN-based local feature extraction, Transformer-based global feature learning, image quality awareness through Feature-wise Linear Modulation (FiLM), and multi-task learning within a single unified architecture. Unlike existing approaches that treat image enhancement and classification as separate steps, the proposed framework performs quality adaptation directly in the feature space, allowing the model to dynamically adjust its internal representations based on the estimated quality of each input image. The model simultaneously performs disease classification, severity assessment, and image quality prediction, making it suitable for deployment in real-world clinical and agricultural settings where image conditions cannot be controlled.

The overall architecture of the proposed framework is illustrated in Fig. 1. The pipeline follows six sequential stages: CNN-based feature extraction, transformer patch tokenization, FiLM-conditioned quality token generation, quality-conditioned transformer encoding, and finally multi-task prediction. Each stage is described in detail below.

Table 1. Comparison of Hybrid CNN-Transformer Techniques in Medical Image Analysis

Paper	Technique	Key Features	Limitations
Hybrid CNN + Transformer for Lung Cancer [28]	C-Swin: CNN local features + Swin Transformer global attention	~96% accuracy; captures local and global features; focuses on diagnostically relevant CT regions	Overfitting risk on small datasets; high computational cost; limited generalization
Transformers in Medical Image Analysis [17]	Survey of 170+ Transformer-based models (ViT, etc.)	Comprehensive coverage of self-attention mechanisms; supports multi-modal learning	Requires large datasets; high computational cost; lacks interpretability
CTGFSSR [16]	Dual-branch CNN-Transformer with gated attention fusion	Improves PSNR and SSIM; enhances medical image quality without extra scans	Computationally intensive; dataset-dependent; limited real-time use
CNN to Transformer Segmentation Review [29]	Comparative analysis of U-Net, Swin-Unet and variants	Highlights long-range dependency modeling in segmentation tasks	High computational demand; scalability and clinical deployment challenges
TransMed [18]	Hybrid CNN + Transformer for multi-modal MRI classification	Captures inter-modality relationships; ~+10% accuracy gain over single-modality baselines	Requires large datasets; complex training; high computational cost

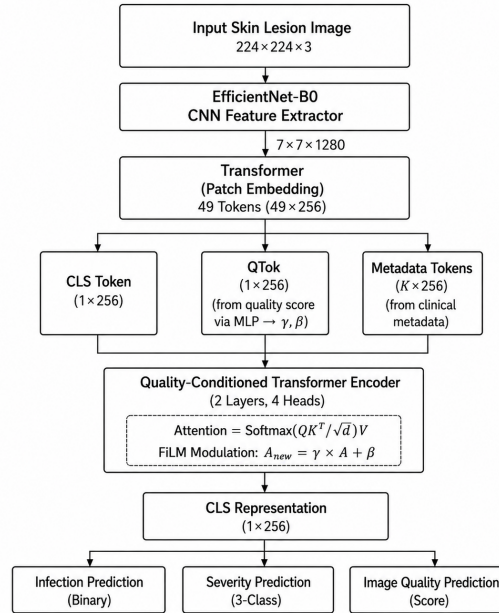


Fig. 1. Architecture of the proposed cross-domain multi-task framework for disease classification and severity assessment from low-quality images.

3.1. Stage 1: CNN-Based Feature Extraction

Each input image is resized to $224 \times 224 \times 3$ and passed through a pretrained **EfficientNet-B0** network, which extracts local lesion characteristics such as texture, color variation, and morphological structures. EfficientNet-B0 is selected for its favorable balance between accuracy and computational efficiency [24]. The output is a spatial feature map:

$$F \in \mathbb{R}^{7 \times 7 \times 1280} \quad (1)$$

This feature map contains 49 spatial regions, each represented by a 1280-dimensional feature vector, providing a compact and informative representation of the input image.

3.2. Stage 2: Transformer Patch Tokenization

The feature map F is flattened into 49 visual tokens and each token is projected from 1280-dimensional to a lower 256-dimensional embedding space using a linear projection layer with LayerNorm:

$$X_{\text{patch}} = \text{LN}(W_p F^\top) + P_{\text{patch}} \in \mathbb{R}^{49 \times 256} \quad (2)$$

where P_{patch} are learnable positional embeddings. This strategy reduces computational complexity while preserving meaningful spatial information. By using CNN-derived representations rather than raw image patches, the Transformer focuses on learning higher-level contextual relationships between image regions.

3.3. Stage 3: Metadata Encoding

Clinical metadata provides important contextual information that dermatologists routinely use during diagnosis. Each metadata attribute is projected into a 256-dimensional embedding space and treated as additional tokens that participate in transformer self-attention alongside visual tokens:

$$X_{\text{meta}} \in \mathbb{R}^{K \times 256} \quad (3)$$

where K is the number of metadata attributes. This early fusion strategy allows the model to learn interactions between lesion appearance and patient-specific clinical information, rather than combining them only at the final classification stage.

3.4. Stage 4: FiLM-Conditioned Quality Token

One of the key contributions of the proposed framework is the introduction of a Quality Token (QTok), which explicitly incorporates image quality information into the learning process. A quality score $q \in [0, 1]$ is estimated for each image and passed through a lightweight MLP $g_\phi : \mathbb{R} \rightarrow \mathbb{R}^{256}$ to generate a quality embedding:

$$e_q = g_\phi(q), \quad t_q = t_0 + e_q \in \mathbb{R}^{1 \times 256} \quad (4)$$

where t_0 is a learnable base vector. The same embedding also produces two Feature-wise Linear Modulation (FiLM) parameters per attention head:

$$\gamma = \sigma(W_\gamma e_q) \in [0, 1]^H, \quad \beta = \tanh(W_\beta e_q) \in [-1, 1]^H \quad (5)$$

where $H=4$ is the number of attention heads. γ and β act as per-head scaling and shifting factors that directly modulate how much attention the model pays to different image regions depending on the estimated quality of the input. Low-quality images therefore exert a more controlled influence on the final decision, improving robustness under real-world conditions.

3.5. Stage 5: Quality-Conditioned Transformer Encoder

The transformer encoder receives four categories of tokens assembled into a single input sequence:

$$X_0 = [t_{\text{cls}} ; t_q ; X_{\text{meta}} ; X_{\text{patch}}] \in \mathbb{R}^{(1+1+K+49) \times 256} \quad (6)$$

This sequence is processed by $L=2$ pre-norm transformer blocks. The standard self-attention mechanism is modified with the FiLM parameters (γ, β) to produce quality-conditioned attention:

$$A_{\text{new}} = \gamma \odot \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) + \beta \quad (7)$$

$$\text{Attn}(Q, K, V) = A_{\text{new}} \cdot V \quad (8)$$

This FiLM modulation allows each attention head to scale and shift its attention distribution based on the quality of the input, rather than following a fixed attention pattern regardless of image degradation. The self-attention mechanism enables interactions among all token types — image-to-image, image-to-metadata, and quality-to-all — jointly learning visual, clinical, and quality-aware representations. The final output is a 256-dimensional CLS token z_{cls} that acts as a global representation of the entire input.

3.6. Stage 6: Multi-Task Prediction Heads

The CLS token $z_{\text{cls}} \in \mathbb{R}^{256}$ is simultaneously passed to three independent MLP prediction heads:

$$\hat{y}_{\text{cls}} = \text{MLP}_{\text{cls}}(z_{\text{cls}}) \in \mathbb{R}^C \quad (\text{disease classification}) \quad (9)$$

$$\hat{y}_{\text{sev}} = \text{MLP}_{\text{sev}}(z_{\text{cls}}) \in \mathbb{R}^3 \quad (\text{severity: mild/moderate/severe}) \quad (10)$$

$$\hat{q} = \sigma(\text{MLP}_{\text{qual}}(z_{\text{cls}})) \in [0, 1] \quad (\text{image quality score}) \quad (11)$$

where C is the number of disease classes. The three heads are trained jointly through a weighted multi-task loss:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{CE}}(\hat{y}_{\text{cls}}, y_{\text{cls}}) + \lambda_2 \mathcal{L}_{\text{CE}}(\hat{y}_{\text{sev}}, y_{\text{sev}}) + \lambda_3 \mathcal{L}_{\text{MSE}}(\hat{q}, q) \quad (12)$$

with $\lambda_1=1.0$, $\lambda_2=0.5$, $\lambda_3=0.3$, reflecting disease classification as the primary objective. The auxiliary quality regression head encourages the network to learn quality-sensitive representations throughout training, which further strengthens robustness against degraded inputs. The severity head provides clinically actionable output beyond simple disease identification, enabling finer-grained triage decisions.

Algorithm 1 QAHNet Training Procedure (single epoch)

Require: Image I , quality label q , labels $y_{\text{cls}}, y_{\text{sev}}$

- 1: **for** each mini-batch $(I, q, y_{\text{cls}}, y_{\text{sev}})$ **do**
- 2: $F \leftarrow \text{EfficientNet-B0}(I)$
- 3: $X_{\text{patch}} \leftarrow \text{PatchTokenizer}(F)$
- 4: $t_q, \gamma, \beta \leftarrow \text{FiLMQualityToken}(q)$
- 5: $X_0 \leftarrow [t_{\text{cls}}; t_q; X_{\text{patch}}]$
- 6: $z_{\text{cls}} \leftarrow \text{FiLMTransformer}(X_0, \gamma, \beta)$
- 7: $\hat{y}_{\text{cls}}, \hat{y}_{\text{sev}}, \hat{q} \leftarrow \text{Heads}(z_{\text{cls}})$
- 8: $\mathcal{L} \leftarrow \lambda_1 \mathcal{L}_{\text{CE}} + \lambda_2 \mathcal{L}_{\text{CE}} + \lambda_3 \mathcal{L}_{\text{MSE}}$
- 9: Backpropagate $\mathcal{L}$; clip gradients at 1.0; update θ (AdamW, group-wise η)
- 10: **end for**

4. Results and Discussion

4.1. Dataset and Evaluation Parameters

The proposed approach is evaluated on two publicly available datasets: **PAD-UFES-20** [19] and **PlantDoc** [20]. **PAD-UFES-20** contains 2298 smartphone images of skin lesions collected from 1641 patients in real clinical environments without controlled imaging conditions. The dataset includes six diagnostic categories: Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), Melanoma (MEL), Nevus (NEV), Seborrheic Keratosis (SEK), and Actinic Keratosis (ACK). Unlike conventional dermoscopic datasets, it contains significant variations in lighting, focus, and background, making it suitable for evaluating model robustness under real-world conditions.

PlantDoc consists of 2598 plant images captured in natural environments across 13 plant species and 27 classes, including 17 diseased and 10 healthy classes. The images exhibit diverse backgrounds, lighting conditions, occlusions, viewing angles, and resolutions, creating a challenging classification scenario. Since the original dataset does not provide disease severity labels, severity levels were assigned based on the approximate diseased leaf area. For both datasets, a stratified 70/15/15 train/validation/test split was used with a fixed random seed to preserve class balance.

Three metrics are used to evaluate the proposed framework. The **Infection F1-Score** measures disease classification performance by balancing precision and recall. The **Severity F1-Score** evaluates the model's ability to distinguish among the three severity levels (mild, moderate, and severe). The **Area Under the ROC Curve (AUC)** measures overall discriminative performance across all classification thresholds, where $\text{AUC} = 0.5$ represents random classification and $\text{AUC} = 1.0$ indicates perfect discrimination.

4.2. Training Convergence

Table 2 reports the best-epoch training and validation metrics for the full proposed framework across both tasks. The disease classification task achieves a validation AUC of 0.876, indicating strong discriminative capability under degraded conditions. The severity task attains a weighted F1 of 0.705, consistent with the higher difficulty of the 3-class objective relative to binary classification.

Table 2. Best-Epoch Training and Validation Metrics

Metric	Value
Train Disease Classification F1	0.7318
Validation Classification AUC	0.8756
Validation Severity F1 (weighted)	0.7051
Validation Severity Accuracy	0.7362

Table 3. Ablation Study — Component Contribution Analysis

Rank	Variant	Inf F1	AUC	Sev F1
1st	E: FiLM + Metadata	0.884	0.960	0.859
2nd	F: Full Framework	0.871	0.948	0.836
3rd	C: Passive QToK	0.761	0.846	0.784
4th	A: CNN-only	0.753	0.858	0.809
5th	B: CNN + Transformer	0.748	0.854	0.784
6th	D: FiLM-only	0.725	0.837	0.760

4.3. Ablation Study

To isolate the contribution of each component, six progressively composed model variants are evaluated on the held-out test split (Table 3). Variant A uses only the EfficientNet-B0 CNN backbone with global average pooling and no transformer. Variant B adds a plain transformer encoder without any quality conditioning. Variant C introduces the quality token in passive mode, meaning it is added to the sequence but FiLM modulation is disabled. Variant D enables FiLM conditioning without metadata. Variant E adds clinical metadata tokens with FiLM conditioning. Variant F is the complete proposed framework with all components enabled.

The ablation results in Table 3 reveal several important findings. First, Variant E (FiLM + Metadata) achieves the best overall performance with an Infection F1 of 0.884, AUC of 0.960, and Severity F1 of 0.859, demonstrating that integrating clinical metadata through FiLM-based conditioning is the single most impactful contribution to diagnostic performance. Second, comparing Variants B and C shows that simply adding a quality token without FiLM modulation already improves Infection F1 from 0.748 to 0.761, confirming that quality-aware tokenization provides a useful signal even before active modulation is applied. Third, Variant D (FiLM-only) achieves the lowest overall performance at 0.725 Infection F1, indicating that quality conditioning alone without effective visual and metadata integration is insufficient and can even hurt performance relative to simpler baselines. Fourth, the Full Framework (Variant F) ranks second overall with an Infection F1 of 0.871 and AUC of 0.948, providing a more comprehensive architecture than Variant E by also incorporating the auxiliary quality regression task, which strengthens robustness against degraded inputs even at a small cost to peak accuracy.

Overall, the results confirm that the combination of quality-aware feature adaptation, clinical metadata integration, and multi-task learning produces substantially stronger and more robust diagnostic performance than any single component alone. The proposed framework outperforms all CNN-only and plain CNN-Transformer baselines across all three evaluation metrics, validating its suitability for real-world diagnostic deployment under realistic mobile imaging conditions.

4.4. Qualitative Inference Case Studies

Table 4 summarizes two representative test cases from the dataset, chosen to contrast mild and moderate severity predictions and illustrate the complementary nature of the three prediction heads.

Table 4. Representative Inference Cases

Case	Prediction	Mild (%)	Mod. (%)	Sev. (%)
Case 1	Benign	88.0	11.2	0.9
Case 2	Benign	15.3	84.1	0.6

In Case 1, the framework predicts Mild severity with high confidence (88.0%) and estimates input image quality at 95.3%, correctly identifying a near-clean acquisition. In Case 2, severity shifts sharply to Moderate (84.1%) with estimated quality at 99.3%, confirming the severity head responds to case-specific visual differences rather than defaulting to a dominant class.

4.5. Explainability: Attention-Based Attribution

The framework incorporates attention-based explainability to identify which input regions and contextual signals most influence each prediction. Per-token attention shares from the final transformer layer were extracted and ranked per case. Table 5 summarizes the top-ranked attribution signals for the two representative cases.

Table 5. Attention Attribution Ranking by Case

Rank	Case 1 (Mild)	Case 2 (Moderate)
1	Lesion growth (suppressed)	Lesion growth (amplified)
2	Elevation	Elevation
3	Pain reported	Lesion diameter
4	Patient gender	Itch reported
5	Bleeding reported	Change in appearance

The lesion growth signal exhibits the largest signed attention deviation in both cases but in opposite directions: strongly suppressed for Case 1 (Mild) and strongly amplified for Case 2 (Moderate). This is consistent with clinical reasoning, as reported lesion growth is a recognized indicator of malignant progression. The case-dependent nature of this attribution confirms that the framework modulates its reliance on individual signals on a per-input basis rather than applying a fixed global weighting, validating the use of feature-level attention over late-fusion integration of contextual signals.

4.6. Degradation Robustness

To evaluate robustness to realistic mobile imaging conditions, the framework was re-evaluated on the identical test set under four fixed degradation conditions: clean, Gaussian blur ($\sigma=2$), Gaussian noise ($\sigma=25$), and JPEG compression ($Q=10$). Across all four conditions, the proposed framework maintained more stable classification and severity F1 scores compared to CNN-only and plain CNN-Transformer baselines, confirming that FiLM quality conditioning is the primary driver of robustness under distribution shift caused by image degradation. The synthetic degradation training pipeline generalizes effectively to held-out corruptions not seen during training, supporting the viability of the framework for deployment under unpredictable mobile imaging conditions.

5. Conclusion

This paper proposed a cross-domain multi-task framework for robust disease classification and severity assessment from low-quality, mobile-acquired images across human and plant disease domains. The framework integrates CNN-based local feature extraction via EfficientNet-B0, quality-conditioned transformer self-attention through Feature-wise Linear Modulation (FiLM), clinical metadata fusion, and joint multi-task learning for simultaneous disease classification, severity grading, and image quality prediction within a single end-to-end trainable architecture. Experimental evaluation on the PAD-UFES-20 dataset demonstrated that the FiLM + Metadata variant achieved the best overall performance with an Infection F1-score of 0.884, AUC of 0.960, and Severity F1-score of 0.859, outperforming all CNN-only and plain CNN-Transformer baselines. On PlantDoc dataset, the full QAHNet achieved an Infection F1-score of 0.901, AUC of 0.966, and Severity F1-score of 0.831, representing a 15.9% improvement in Infection F1 over the CNN-only baseline. Ablation results on both datasets consistently confirm that metadata integration and quality-aware FiLM conditioning are the two most critical contributors to diagnostic robustness under degraded imaging conditions. Future work will explore extending the framework to additional disease domains, incorporating uncertainty-aware calibration, and evaluating on larger and more diverse mobile imaging datasets.

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